

Government Engineering College, Thrissur Department of Electronics and Communication Engineering					
PROPOSAL FOR SEMINAR					
Course Code	ECQ413	Semester	VII	Program	BTECH - ECE
Course Name	SEMINAR	Batch	2022 - 2026	Proposal No.	
Reg. No.	TCR22EC066	Name of the student		VISHNU DIVAKAR	
Title of Seminar	Wi-Fi 7 : Unlocking the Future of High-Speed and Low-Latency Wireless Communication				
Relevance of the topic	Enables ultra-fast, low-latency wireless communication with advanced modulation, AI-driven optimization, and support for real-time, high-demand applications.	Technical Contents		Multi-Link Operation, 4096-QAM, AI-based channel selection, TSN support, Real-time IoT/AR applications	
Name of the guide	Dr. JOB CHUNKATH			Accept	Reject
<i>Give an Abstract of the Seminar limited to 200 – 300 words</i>					
Abstract					
Introduction to Wi-Fi 7 • Wi-Fi 7 (IEEE 802.11be) is the latest evolution in wireless LAN standards. • Aims to deliver extremely high throughput, ultra-low latency, and improved reliability. Key Technological Features • Multi-Link Operation (MLO): Combines multiple frequency bands (2.4 GHz, 5 GHz, and 6 GHz). • 320 MHz channel bandwidth and 4096-QAM for enhanced data capacity. • 16 spatial streams with advanced MU-MIMO techniques. • TSN (Time-Sensitive Networking) support for mission-critical communications. AI and Machine Learning Integration • Use of reinforcement learning and bandit algorithms for dynamic channel selection. • Real-time traffic allocation and interference management in dense networks. Experimental and Real-World Validation • Validation on commercial Wi-Fi 7 chipsets for packet-level steering. • Research shows 30–50% reduction in latency and increased throughput up to 30 Gbps. Applications and Relevance to ECE • Applications in cloud gaming, 8K video streaming, AR/VR, and industrial IoT. • Supports real-time control systems in smart factories and healthcare. • Opens opportunities in embedded systems, network optimization, and RF design.					

Conclusion

- Wi-Fi 7 is a major leap in wireless communication that blends cutting-edge hardware with AI-driven intelligence.
- It promises to redefine the landscape of wireless connectivity for next-gen applications.

References *(Give references in IEEE format)*

1. S. Lian et al., "Intelligent Channel Allocation for IEEE 802.11be Multi-Link Operation," arXiv preprint arXiv:2506.04594, 2025.
2. Y. Liu et al., "IEEE 802.11be Wi-Fi 7: Feature Summary and Performance Evaluation," arXiv:2309.15951, 2024.
3. M. Rosani et al., "A Software Platform for Testing Multi-Link Operation," arXiv:2411.12077, 2024.
4. G. Cena et al., "Packet Steering Mechanisms for MLO in Wi-Fi 7," arXiv:2411.13470, 2024.
5. A. Iturria-Rivera et al., "Multi-Headed RL for Traffic Allocation," arXiv:2303.08959, 2023.

Signature of Guide: